SMS Spam Detection: NLP-Based Classification System

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Project Objectives

Security Enhancement Objective

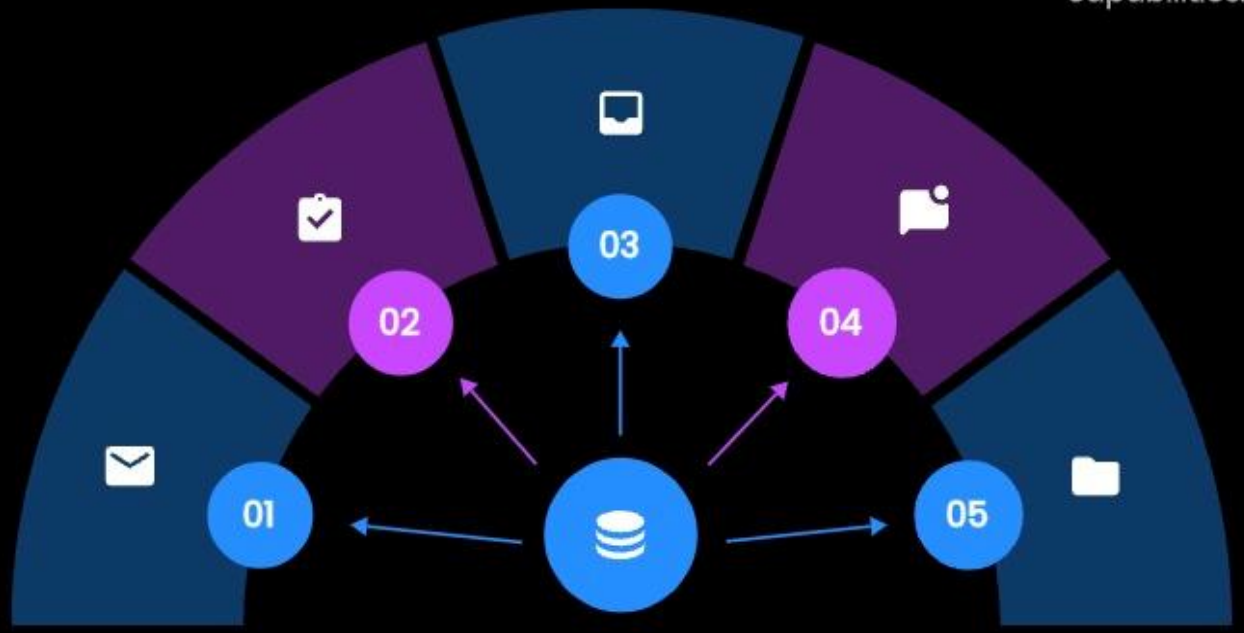
Improve user security and reduce fraud risks through intelligent message filtering systems.

Key Classification Objective

Classify SMS messages as spam or ham with high precision and reliability for accurate detection.

Application Development Objective

Build practical web and mobile applications enabling end-users real-time spam detection capabilities.



Primary Goal

Detect spam messages using machine learning algorithms to protect user communications effectively.

Expected Outcome

Achieve real-time spam detection with high accuracy and reliable performance metrics.

Stakeholder Analysis

Mobile Users

Direct beneficiaries seeking reliable spam protection and enhanced messaging security experience.

Telecom Companies

Providers implementing fraud prevention systems to reduce financial losses and improve service quality.

Cyber Security Teams

System monitoring and threat analysis professionals ensuring platform safety and detecting emerging risks.

Application Developers

Implementation and maintenance support specialists responsible for system integration and technical updates.

Dataset & Model

Dataset : The SMS Spam Collection Dataset contains 5,574 labeled messages categorized as spam or ham. Each entry includes a label and the corresponding message text. Preprocessing steps such as tokenization, stopword removal, and TF-IDF vectorization are applied.

Model : A Multinomial Naïve Bayes classifier is used due to its efficiency and effectiveness in text classification. The model predicts whether a message is spam or ham based on word probabilities and is evaluated using accuracy, precision, recall, and F1-score.

9. Test with User Input

[11]
✓ 12s

```python
# Function to predict new message
def predict_spam(message):
    msg_tfidf = tfidf.transform([message])
    prediction = model.predict(msg_tfidf)[0]

    if prediction == 1:
        return "Spam"
    else:
        return "Not Spam"

# User input
user_msg = input("Enter SMS: ")
print("Prediction:", predict_spam(user_msg))
```

```
Enter SMS: helooo
Prediction: Not Spam
```

9. Test with User Input

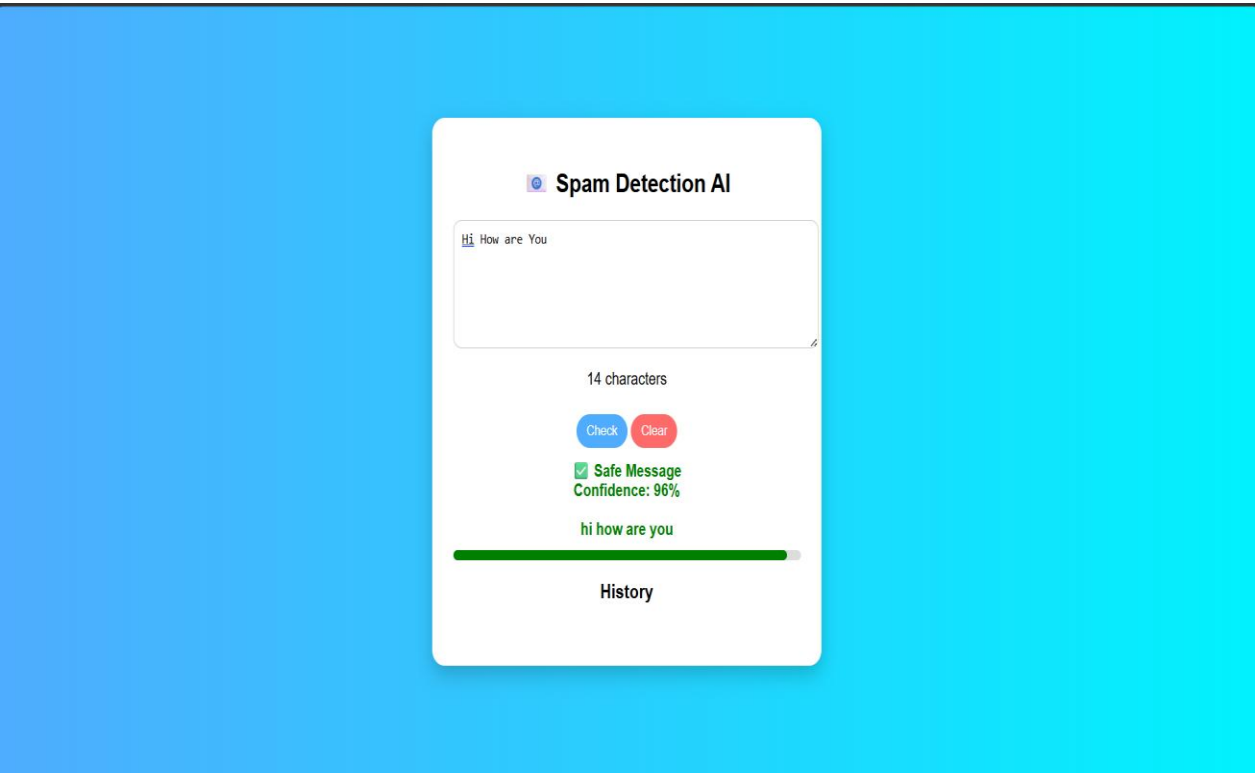
[12]
✓ 5s

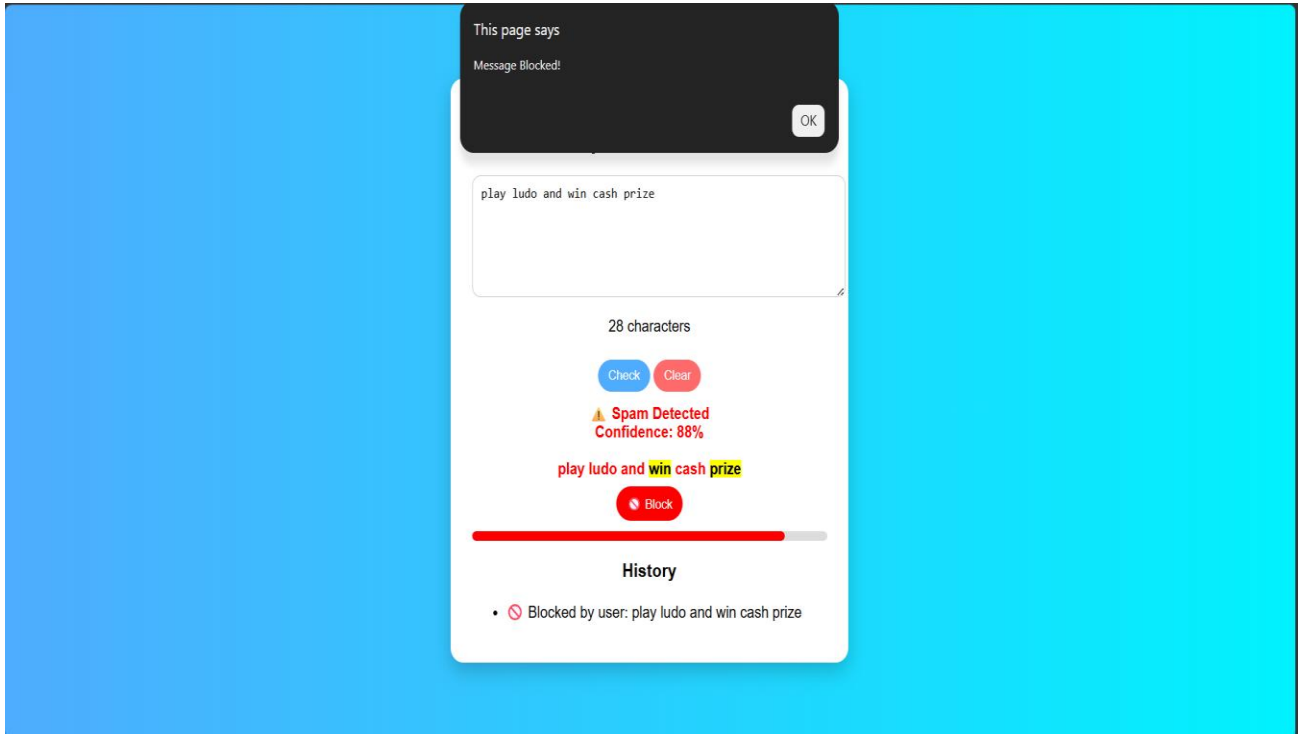
```python
# Function to predict new message
def predict_spam(message):
    msg_tfidf = tfidf.transform([message])
    prediction = model.predict(msg_tfidf)[0]

    if prediction == 1:
        return "Spam"
    else:
        return "Not Spam"

# User input
user_msg = input("Enter SMS: ")
print("Prediction:", predict_spam(user_msg))
```

```
Enter SMS: urgent
Prediction: Spam
```


Spam Detection AI
Hi How are You
14 characters
Check
Clear
Safe Message
Confidence: 96%
hi how are you
History


This page says
Message Blocked!
OK
play ludo and win cash prize
28 characters
Check
Clear
Spam Detected
Confidence: 88%
play ludo and win cash prize
Block
History
Blocked by user: play ludo and win cash prize

Application

📧 Email Services: Automatically filters spam and phishing emails in platforms like Gmail and Outlook.

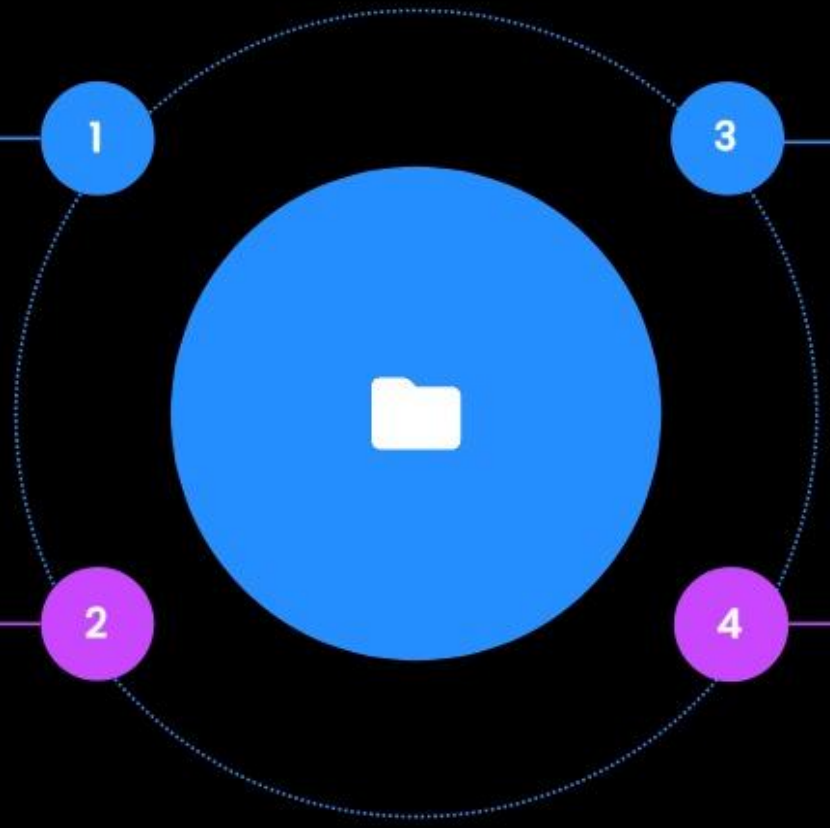
🏢 Enterprise Security: Protects organizational email systems from malicious and unwanted messages.

🌐 Web Applications: Prevents spam in contact forms and user registrations.

📱 Mobile & SMS Platforms: Detects and blocks spam messages in messaging apps.

🛍️ E-commerce Platforms: Filters fraudulent promotions and fake communications.

Conclusion & Next Steps



1

Project Achievement

Successfully developed ML-based spam detection system with robust classification capabilities and comprehensive performance validation across datasets.

2

Deployment Impact

Enhanced user security and fraud prevention while reducing spam incidents significantly, improving overall platform reliability and customer trust.

3

Key Recommendations

Gather user feedback for feature improvements, integrate with telecom platforms, and implement deep learning models for enhanced accuracy.

4

Implementation Timeline

Full system deployment expected within three to six months with phased rollout across major platforms and continuous monitoring.

References

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UCI Machine Learning Repository – SMS Spam Collection.

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McCallum, A. & Nigam, K. (1998). *Naïve Bayes for Text Classification*.

AWS, Google Cloud, and Microsoft Azure – Official Documentation.

Thank you!